

SYNOPSIS



On 20 June 2001, the 8,904gt Greek-registered ro-ro cargo vessel *Thelisis*, collided with the 14.02m UK-registered fishing vessel *Our Sarah Jayne* in the Thames Estuary. The accident was reported to the Marine Accident Investigation Branch (MAIB) that day, and an investigation began immediately.

At the time of the collision *Thelisis* was under pilotage and *Our Sarah Jayne* was engaged in fishing.

As a result of the collision, *Our Sarah Jayne* flooded then foundered. The skipper, who was operating his vessel single-handedly, was rescued by a Thames pilot cutter. There were no injuries. *Thelisis* sustained slight damage to her hull plating.

The cause of the accident was the failure by the pilot on board *Thelisis* to take avoiding action in sufficient time, and the failure of the skipper on board *Our Sarah Jayne* to keep a proper lookout in accordance with the *Collision Regulations*.

Contributing factors were:

- The pilot's reliance on the use of VHF radio for collision avoidance.
- The pilot's complacency and expectation that *Our Sarah Jayne* would alter course on his request.
- The master of *Thelisis* failing to override the pilot's instructions.
- The decision by *Our Sarah Jayne*'s skipper to sail single-handedly, thereby denying himself the ability to keep a safe navigational watch.

Recommendations have been addressed to the PLA, the owner of *Thelisis* and also to *Our Sarah Jayne*'s skipper and owner, which if implemented should reduce the risk of such an accident recurring.

SECTION 1 - FACTUAL INFORMATION

(All times are UTC + 1)

1.1 PARTICULARS OF THELISIS/OUR SARAH JAYNE AND ACCIDENT

Vessel details *Thelisis*

Registered owner	:	Bonanza Maritime, Piraeus, Greece
Manager	:	Efthymiou Shipping, Piraeus, Greece
Port of registry	:	Piraeus
Flag	:	Greece
Type	:	Ro-ro cargo
Built	:	1979
Classification society	:	Registro Italiano Navale
Construction	:	Steel
Length overall	:	132.8m
Breadth	:	20.32m
Gross tonnage	:	8904
Engine power and type	:	5370kW Heavy oil
Service speed	:	18 knots

Accident details

Time and date	:	0119 on 20 June 2001
Location of incident	:	Thames Estuary
People on board	:	19
Injuries/fatalities	:	None
Damage	:	Slight shell plate damage – aft starboard quarter

Vessel details***Our Sarah Jayne***

Registered owner	:	Mr B J Martin, Whitstable, Kent
Port of registry	:	Brixham
Type	:	Trawler
Flag	:	UK
Fishing Number	:	BM 116
Built	:	1973 Appledore, Devon
Construction	:	Wood
Length overall	:	14.02m
Registered length	:	13.42m
Breadth	:	4.88m
Depth	:	1.68m
Gross tonnage	:	20.94
Engine power and type	:	130kW
Service speed	:	10 knots

Accident details

Time and date	:	0119 20 June 2001
Location of incident	:	Thames Estuary
Position of foundering	:	51° 28.6' N 001° 23.9' E
People on board	:	1
Injuries/fatalities	:	None
Damage	:	Total loss

1.6 THE PILOT

The pilot was an experienced mariner.

Before his employment with the Port of London Authority (PLA) he had spent several years at sea, first as an officer cadet, and eventually progressing to master. He had served on various vessels worldwide.

He joined the PLA in 1998 as a trainee pilot. Since then he had progressed to a class 2 pilot, and had completed over 350 acts of pilotage.

Normally working a roster system of 9 days on and 6 days off, he was called in on an overtime basis from his off-duty period to conduct the pilotage on board *Thelisis*. His previous pilotage act had been 2 days before the accident.

1.7 ENVIRONMENTAL CONDITIONS

The weather reported throughout the incident was a southerly wind of force 2 with a negligible sea swell. The visibility was very good.

The predicted time of high water at Sheerness on 19 June was 2306 with low water on 20 June at 0523. The ebb tide was flowing in a predicted direction of 067° at a rate of 1.2 knots.

1.8 NARRATIVE OF EVENTS (ALL COURSES ARE TRUE)

Thelisis arrived at the NE Spit pilot station, east of Margate Road, and dropped anchor at 1800 on 19 July 2001. The pilot was ordered for later on that evening for passage to Tilbury dock, where she was due to arrive at 0600 the following morning to discharge her cargo.

At 1900 *Our Sarah Jayne*'s skipper left home and made his way to the harbour to join his vessel. He had been at home, asleep all afternoon, as he usually did when engaged in night fishing. Arriving at the vessel at 2015, he loaded some stores and made preparations for sailing. At 2045, in the company of the fishing vessels *Charlie Boy* and *Tiscino*, *Our Sarah Jayne* left Whitstable harbour bound for the fishing grounds close to Margate Road. Shortly before 2300, she arrived on the grounds, shot her gear, and began towing in a north-easterly direction.

At 2300, the pilot assigned to *Thelisis* left Gravesend by taxi, arriving an hour later in Ramsgate where he joined the pilot cutter. At approximately the same time *Our Sarah Jayne* hauled her gear; the catch was poor so the skipper decided to steam north to Queens Channel. Arriving at 0030, she once again shot her gear and this time began towing on an approximate course of 270° at a speed of about 2 knots through the water.

As the pilot boarded *Thelisis* her crew were in the process of weighing anchor. The time was 0050.

Once on board, the pilot was escorted to the bridge and introduced to the master. The second officer and a helmsman were also on the bridge. The pilot explained the passage plan to the master, had it signed, asked some questions relevant to the pilot card and then ordered half speed.

On board *Our Sarah Jayne*, the skipper was on watch sitting in the wheelhouse chair. The radar was switched off and the vessel was being steered by auto-pilot. One VHF radio was tuned to channel 12 and the other to channel 6. The skipper observed three outward bound vessels passing to the north. He was also aware of another vessel, inbound, some distance to the south-east, which he considered would pass astern.

After rounding Margate buoy the pilot on *Thelisis*, which was in manual steering with a dedicated helmsman, ordered a course of 290°. The second officer was manning the engine controls, and both the master and the pilot were looking out ahead.

The pilot had decided to steer a course between three vessels to the north-west of *Thelisis*, outward bound in the Princes Channel, and another three vessels to the south, from the Medway ports, who were, or shortly would be, landing pilots.

When *Thelisis* came on to a course of 290° her speed was approximately 8 to 9 knots. Both the pilot and the master then noticed a white light ahead at an approximate range of 1.5 miles. They were both unsure as to the nature of the light, so the pilot called VTS to ask if they could provide any information on it from their radar coverage. The time was 0108. VTS replied to the effect that they did not know the name of the vessel and said "He's not doing anything. He's zero".

Because of the outward bound traffic, the pilot decided to pass to the south of the vessel, although there was sufficient searoom to pass on either side. He altered course to port by a few degrees and, shortly afterwards, became aware both visually and by radar, that the other vessel was making way and on a steady bearing with the distance decreasing. The range of the other vessel was then approximately 1 mile. He tried calling her twice on VHF radio channel 12 at 0113. The only reply to the VHF radio calls came at 0117 from *Spruce*, an outward bound vessel in the Princes Channel. Initially, the pilot thought *Spruce* was the name of the fishing vessel ahead; however after a short time he realised this was not the case. He asked the second officer to sound prolonged blasts on the whistle, and to direct the searchlight to attract the other vessel's attention. Realising the imminent danger of a collision, the pilot then ordered an alteration of course hard to port. As the stern of *Thelisis* swung round to starboard, her starboard quarter collided with *Our Sarah Jayne*'s port side. The time was 0119.

Our Sarah Jayne's skipper first became aware of *Thelisis* in close proximity when he noticed her wash out of a port side wheelhouse window. The next thing he recalled was being virtually thrown from the wheelhouse chair when the vessels collided. As a result of the collision *Our Sarah Jayne* was inclined to such an extent that seawater started to pour through a partly open wheelhouse window. In addition to this, alarms began to sound in the wheelhouse as seawater entered the engine room through the damaged hull.

The skipper tried to contact *Thelisis* using VHF radio channel 12, but was unsuccessful. He then tried to make contact first with VTS, and then the coastguard on the same channel, before he managed eventually to contact the pilot cutter on channel 9 and report his situation. This was followed by another call soon after; this time reporting that the main engine had stopped and the vessel was flooding rapidly. The pilot cutter relayed this information to VTS and the pilot station who, in turn, informed the coastguard.

The pilot on board *Thelisis* was unaware that the vessels had collided and continued the port manoeuvre, eventually making a full turn to port before swinging back to starboard, clearing *Our Sarah Jayne* to the south. He then heard on the VHF radio that the pilot cutter was on her way to assist the fishing vessel. On hearing this he ordered dead slow and informed the pilot cutter and VTS that he would be standing by, in case further assistance was required.

As soon as the pilot cutter arrived alongside *Our Sarah Jayne*, the skipper abandoned his vessel, which was by now heavily listed to starboard because of the floodwater. Twenty minutes later *Our Sarah Jayne* sank.

The skipper was transferred ashore at Ramsgate by the pilot cutter, and *Thelisis* continued her passage upriver.

1.9 THAMES VTS

Thames VTS provides a navigation service for vessels entering and leaving the area under the control of the Port of London Authority (PLA).

A station situated in Gravesend provides coverage for the lower region, including the Thames estuaries. It is manned 24 hours a day, monitoring and providing advice to traffic transiting the Thames. All vessels over 40m overall length are required to report to VTS at designated points with their intended movements. Vessels of more than 20m in length are required to maintain a VHF radio listening watch. Neither requirement applied to *Our Sarah Jayne*, since she was 14.02m in length.

Apart from the VTS operators, a duty port controller, who is a Thames class 1 pilot employed on a rotational basis, is responsible for the day to day operation. VTS operators have a good working relationship with the pilots. Navigational advice is given in response to requests from vessels and where safety considerations require. Generally, pilots are aware of the limitations of the advice provided.

The position of the accident is within the coverage area provided by Thames VTS. The events leading up to the accident were recorded both on radar and VHF radio channel 12 **(Figures 4 & 5) (Annex 1)**.

On 27 July 2001, the harbourmaster (Lower District) issued a memo stating that targets acquired by VTS radars to the east of Crayfordness will display track information of zero speed and zero course unless the speed is greater than approximately 2 knots over the ground **(Annex 2)**.

1.10 COLLISION REGULATIONS

Rule 5 of the *International Regulations for Preventing Collisions at Sea 1972 (Collision Regulations)* states:

Every vessel shall at all times maintain a proper look-out by sight and hearing as well as by all available means appropriate in the prevailing circumstances and conditions so as to make a full appraisal of the situation and of the risk of collision.

Rule 8, in part, states:

- (a) *Any action taken to avoid collision shall, if the circumstances of the case admit, be positive, made in ample time and with due regard to the observance of good seamanship.*
- (b) *Any alteration of course and/or speed to avoid collision shall, if the circumstances of the case admit, be large enough to be readily apparent to another vessel observing visually or by radar; a succession of small alterations of course and/or speed should be avoided.*
- (c) *If there is sufficient sea-room, alteration of course alone may be the most effective action to avoid a close-quarters situation provided that it is made in good time, is substantial and does not result in another close quarter situation.*
- (e) *if necessary to avoid a collision or allow more time to assess the situation, a vessel shall slacken her speed or take all the way off by stopping or reversing her means of propulsion.*

Rule 13 states:

Notwithstanding anything contained in the Rules of part B, sections 1 and 2, any vessel overtaking any other vessel shall keep out of the way of the vessel being overtaken.

- (a) *A vessel shall be deemed to be overtaking when coming up with another vessel from a direction of more than 22.5 degrees abaft her beam, that is, in such a position with reference to the vessel she is overtaking, that at night she would be able to see only the sternlight of that vessel but neither of her sidelights.*

Paragraph 5 states:

Although the practice of using VHF radio as a collision avoidance aid may be resorted to on occasion, especially in pilotage waters, the risks described in this Notice should be clearly understood and the Collision Regulations complied with.

VHF radio channel 12 is assigned for ship-ship and ship-shore communications by vessels navigating the lower reaches of the River Thames and its estuaries. As such, it is also used between pilots and other vessels for reporting their progress and advising their navigational intentions.

The PLA does not discourage the use of VHF communication between ships in circumstances where such use can help to resolve possible ambiguity or reduce risk. However, it does not condone the practice of vessels passing dangerously close to one another in violation of the Collision Regulations.

Passing situations are routinely monitored by VTS, which will intervene, when it considers it safe to do so, and will assist a situation. Any such incident observed by VTS, which is deemed to be dangerous or violate the Collision Regulations, is the subject of a formal investigation and, where warranted, formal admonishment.

1.12 NAVIGATIONAL WATCH ON FISHING VESSELS

Guidance and advice in keeping a navigational watch on fishing vessels is given in *Marine Guidance Note MGN 84 (F)* entitled *Keeping a Safe Navigational Watch on Fishing Vessels*.

Section 1 includes:

Investigations into collisions and groundings involving fishing vessels have continued to show that poor watchkeeping is a major cause. In many cases one or more of the following were important factors:

- (a) *an unqualified or inexperienced man in charge of the watch;*
- (b) *only one man on watch (regardless of whether a watch alarm was fitted);*
- (c) *a poor lookout being kept;*
- (d) *divided command, and;*
- (e) *fatigue*

A competent alert watchkeeper, keeping a proper all round look-out at all times is absolutely essential.

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributing causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents from occurring in the future.

2.2 CAUSE OF THE COLLISION

Both the pilot and the master on *Thelisis* first identified the presence of another vessel at a range of approximately 1.5 miles. When the range had decreased to 1 mile, the pilot was aware that *Thelisis* was on a collision course with the other vessel; this was confirmed both visually and by radar. Although unsure if she was a fishing vessel, he was aware that he was overtaking, therefore making his the give-way vessel in accordance with Rule 13 of the *Collision Regulations*.

At that time there was sufficient searoom and depth of water to pass the other vessel either to the north or to the south. Alternatively, there was sufficient time to reduce speed in accordance with Rule 8 of the *Collision Regulations* to assess the situation more fully. An alteration of course by a few degrees only was totally inadequate and not in keeping with Rule 8. When the pilot did order an alteration of course it was far too late.

There is witness evidence to suggest that there might have been an undue delay in applying hard to port helm following the pilot's order. However, this is not supported by other available witness evidence, and it is the opinion of the MAIB that any short delay in applying hard to port helm, for whatever reason, should not detract from the pilot's failure to take early and substantial avoiding action in accordance with Rule 8. Even if the collision could have been avoided, the pilot's actions would still have resulted in an avoidable close quarters situation.

There are also obligations placed on the stand-on vessel, in this case *Our Sarah Jayne*, to take avoiding action to prevent a collision in accordance with Rule 17 of the *Collision Regulations*. However, the skipper was unaware of the situation until immediately before the vessels collided.

Contrary to Rule 5, the skipper was not maintaining a proper lookout.

2.3 VTS INFORMATION

The pilot's decision to alter course to port by a few degrees only was influenced by his misinterpretation of the information he received from VTS to his request as to the status of the other vessel.

At that time neither the PLA, the pilots, nor the VTS operators were aware that targets acquired by VTS radar, to the east of Crayfordness, would display track information as zero speed and course if the speed of the vessel was less than approximately 2 knots over the ground.

Therefore, the pilot did not appreciate that the other vessel was, in fact, making headway over the ground when he decided to alter course to port by a few degrees only, expecting that he would pass clear.

Advice received from VTS regarding navigation should not be relied upon implicitly, but be treated only as part of the overall information available at that particular time. Full use should be made of the vessel's navigational equipment and bridge personnel.

2.4 ACTION BY THE PILOT

With several years experience as master, and 3 years as a Thames pilot, during which time he carried out over 350 acts of pilotage, the pilot should have been fully conversant with the *Collision Regulations*.

Assuming this was the case, why did he leave it so late before substantially altering course?

VHF radio channel 12 is commonly used between pilots and other vessels for reporting their progress and advising their navigational intentions. However, in this case, the pilot relied on VHF radio communications to determine the course of action to be taken when a risk of collision already existed, thereby deferring his obligations under the Collision Regulations and effectively increasing the risk. This danger is highlighted in MGN167.

In accordance with his regular practice of relying on VHF radio for collision avoidance, the pilot attempted to contact *Our Sarah Jayne* in preference to altering course. He had become complacent, and expected that once contacted, the fishing vessel would alter course at the pilot's request, irrespective of her status.

The pilot remained optimistic that the fishing vessel would respond to the situation, until the very last minute, when he considered he had no option but to alter course; by then it was too late.

Not only is it a dangerous practice to rely on VHF radio communications for collision avoidance, but the expectation that the stand-on vessel will give way to the larger vessel makes it extremely so.

In view of the confined nature of the area and the density of passing traffic, there is a potential for this practice to become more common.

SECTION 3 - CONCLUSIONS

3.1 FINDINGS

Thelisis

1. Both the pilot and the master on *Thelisis* were aware they were on a collision course with another vessel at an approximate range of 1 mile. [2.2]
2. The pilot was aware that *Thelisis* was the overtaking vessel in accordance with the *Collision Regulations* [2.2]
3. At that time there was sufficient searoom and depth of water for *Thelisis* to pass the other vessel either to the north or to the south. [2.2]
4. There was also sufficient time to reduce speed, to more fully assess the situation. [2.2]
5. An alteration of course by a few degrees was totally inadequate. [2.2]
6. Had the pilot acted in accordance with Rule 13, and taken action under Rule 8 of the *Collision Regulations*, the accident could have been avoided. [2.2]
7. At the time of the accident, neither the PLA, the pilots, nor the VTS operators were aware that targets acquired by VTS radar, to the east of Crayfordness, would display track information as zero speed and course if the speed of the vessel was less than approximately 2 knots over the ground.[2.3]
8. The pilot's decision to alter course to port by a few degrees only was influenced by his misinterpretation of the information he received from VTS. [2.3]
9. There was a reliance on VHF radio for collision avoidance. [2.4]
10. The pilot expected the skipper of *Our Sarah Jayne* to alter course on receipt of his VHF radio request. [2.4]
11. Without this expectation the collision could have been avoided.[2.4]
12. A possible adverse working relationship between pilots and fishermen exists in the vicinity of where the accident occurred. [2.5]
13. The master could have intervened at any time, overriding the pilot's instructions. Had he done so, the collision could have been avoided. [2.6]
14. The master accepted the pilot's instructions without question. [2.6]

Our Sarah Jayne

15. *Our Sarah Jayne's* skipper was unaware of any danger until immediately before the collision. [2.7]
16. The skipper should have been alerted by the calls made on VHF radio, and any sound signals and searchlight from *Thelisis*. [2.7]
17. Possible reasons for not being alerted were that he failed to appreciate it was his vessel that was being referred to on the VHF radio, he left the wheelhouse unattended, the engine exhaust and decklights reduced his aural and visual perception, or his state of vigilance was reduced while sitting in the wheelhouse chair. However, the skipper reported that he was on watch in the wheelhouse and awake. [2.7]
18. The skipper of *Our Sarah Jayne* was not keeping a proper lookout in accordance with Rule 5 of the *Collision Regulations*. [2.7]
19. A crew of one is considered insufficient for a vessel the size of *Our Sarah Jayne* and her operational routine. [2.8]
20. A minimum of two crew on board could have significantly reduced the risk of collision. [2.8]

3.2 CAUSE

The cause of the collision between *Thelisis* and *Our Sarah Jayne* was the pilot on board *Thelisis* failing to take avoiding action in sufficient time, and the skipper on board *Our Sarah Jayne* failing to keep a proper lookout in accordance with the *Collision Regulations*.

3.3 CONTRIBUTING FACTORS

1. The pilot relying on VHF radio for collision avoidance.
2. The complacency and expectation of the pilot that *Our Sarah Jayne's* skipper would alter course if he requested him to do so.
3. The master of *Thelisis* failing to override the pilot's instructions.
4. The skipper of *Our Sarah Jayne's* decision to sail single-handedly, thereby denying himself the ability to keep a safe navigational watch.